

Response of First-Order Systems

Transfer Function

*Considering unsteady-state behavior
for mercury-in-glass thermometer*

Placed in a flowing stream of fluid

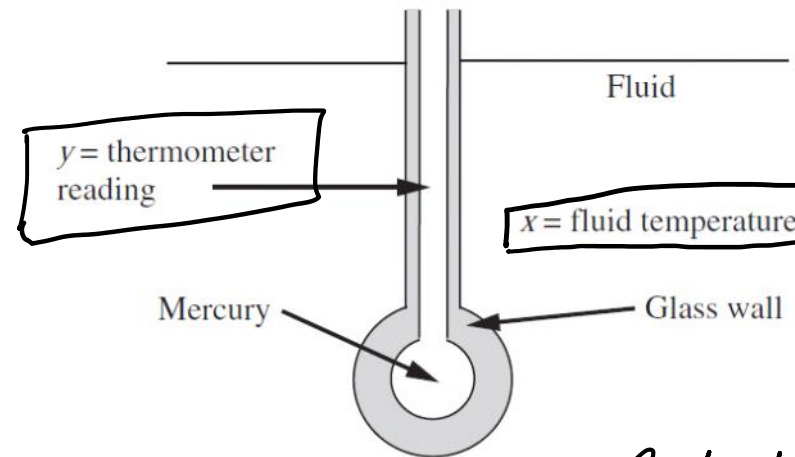


FIGURE 4-1a
Cross-sectional view of thermometer.

** Calculate y as a
function of x*

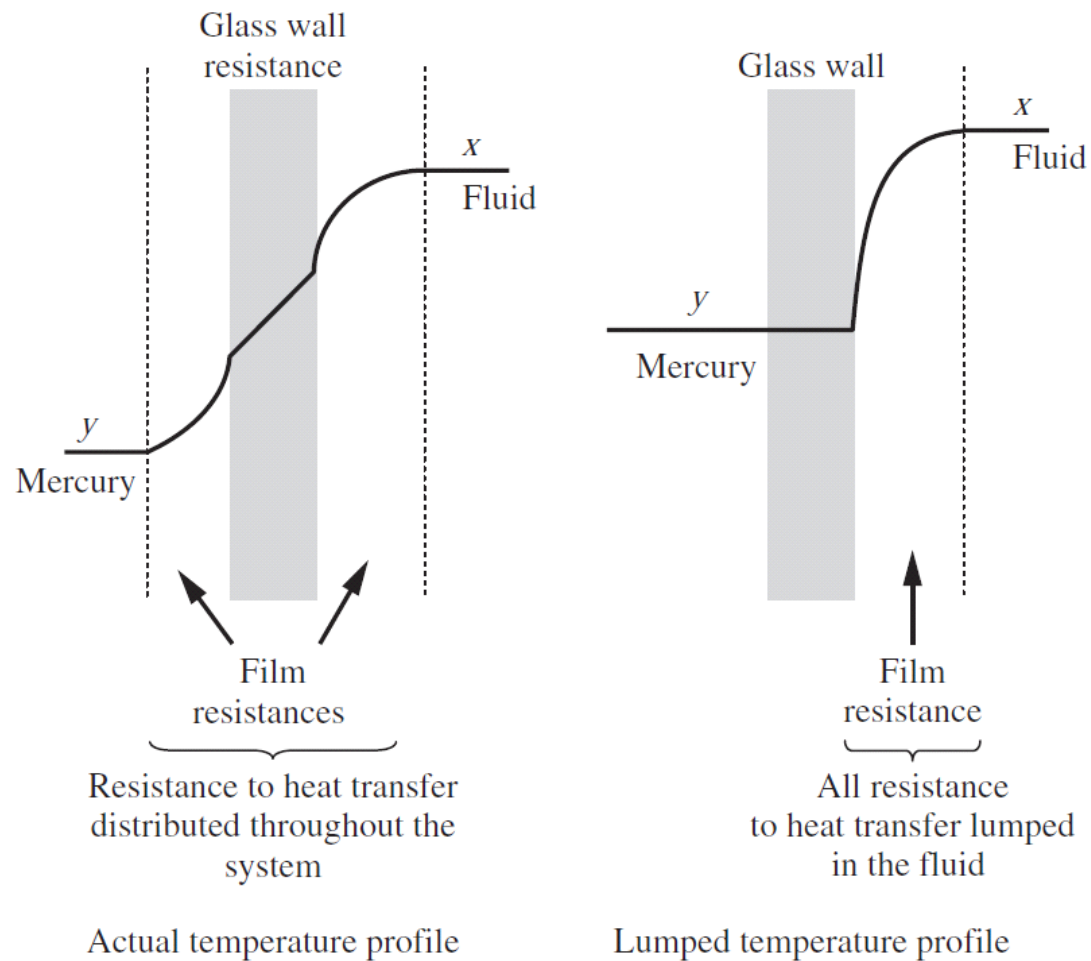


FIGURE 4-1b
Temperature profiles in thermometer.

- ### Analysis Assumptions (Lumping Parameters)
- 1) All heat transfer resistance is present in the film surrounding the bulb
 - 2) Mercury assumes a uniform temperature throughout
 - 3) Glass wall does not expand or contract during transient response
(Real system has inverse response)

Initially Thermometer is at S.S and at time $t=0$; x changes

Applying unsteady state energy balance

$$\text{Input} - \text{Output} = \text{Accumulation}$$

From Newton's Law of Cooling $\dot{Q} = hA (T(t) - T_{\infty}) = hA \Delta T(t)$

$$hA(x-y) - 0 = mC \frac{dy}{dt}$$

} - first order ODE
can be solved using $\mathcal{L}\{f(t)\}$

depends on $\leftarrow$ film
coefficient
of heat transfer
flow rate of
fluid and its
properties

heat capacity
of mercury

Heat input $\rightarrow$ Causes accumulation of energy in Hg and its expansion

Using deviation variables

for eqn $hA(x-y) - 0 = mc \frac{dy}{dt}$ ——— (1)

@ initial s.s. $\frac{dy}{dt} = 0$

Hence @ $t < 0$

$$hA(x_s - y_s) = 0$$

using s for s.s. ——— (2)

where we can only have $x_s = x_y$
meaning that the thermometer reads the true bath temperature

With (1) - (2)

$$hA[(x - x_s) - (y - y_s)] = mc \frac{d(y - y_s)}{dt}$$

$$\Rightarrow x - x_s = X \text{ — input}$$

$$\Rightarrow y - y_s = Y \text{ — output}$$

$$\text{where } \frac{d(y - y_s)}{dt} = \frac{dy}{dt} \text{ as } y_s \text{ is constant}$$

Differential Equation with Deviation Variables

$$hA(X-Y) = mC \frac{dY}{dt}$$

$$X-Y = \frac{mC}{hA} \frac{dY}{dt}$$

$$\boxed{X-Y = \tau \frac{dY}{dt}}$$

$$\begin{aligned}\tau &= \text{Time Constant (time units)} \\ &= \frac{mC}{hA} = \frac{(\text{lbm}) \left(\frac{\text{Btu}}{\text{lbm} \cdot ^\circ\text{F}} \right)}{\left(\frac{\text{Btu}}{\text{ft}^2 \cdot \text{h} \cdot ^\circ\text{F}} \right) (\text{ft}^2)} = \text{h}\end{aligned}$$

Taking Laplace transform

$$\begin{aligned}X(s) - Y(s) &= \tau s Y(s) - Y(0) \\ &= \tau s Y(s)\end{aligned}$$

$$X(s) = \tau s Y(s) + Y(0)$$

$$\begin{aligned}&\rightarrow \text{Steady State} \Rightarrow Y(0) = 0 \\ &= y(0) - y_s \\ &= y_s - y_s = 0\end{aligned}$$

$$X(s) = \tau s Y(s) + Y(s)$$

$$X(s) = Y(s) (\tau s + 1)$$

$$\boxed{\frac{Y(s)}{X(s)} = \frac{1}{\tau s + 1}} = \frac{\text{Output}}{\text{Input}}$$

Transfer
function

Where $\swarrow$
Transfer function
is in this form is called
first-order system

$\Rightarrow$ Ratio of Laplace transforms
of Thermometer temp to
Bath temperature
deviations

Procedure for finding transfer function of a process

- 1) Write mass or energy balance equations and linearize if necessary
- 2) Place balance equations in deviation variable form
- 3) Laplace transform of linear balance equations
- 4) Solve for Output / Input

Standard form for First Order transfer functions

General form for first order systems

$$\tau \frac{dy}{dt} + y = K_p x(t)$$

$y \rightarrow$ output variable

$x(t) \rightarrow$ input forcing function

Initial conditions

$$y(0) = y_s = K_p x(0) = K_p x_s$$

Again using deviation variables

$$X = x - x_s$$

$$Y = y - y_s$$

$$\Rightarrow \tau \frac{dY}{dt} + Y = K_p X(t) \quad ; \quad Y(0) = 0$$

$$\tau \frac{dY}{dt} + Y = K_p X(t)$$

Transforming eqn

$$\tau (s Y(s) - \cancel{Y(0)}) + Y(s) = K_p X(s)$$

$$\tau s Y(s) + Y(s) = K_p X(s)$$

$$\frac{Y(s)}{X(s)} = \frac{K_p}{\tau s + 1}$$

Characteristics of general form

- Denominator of form $\tau s + 1$
- Coefficient of s is system time constant τ
- Numerator is Steady State Gain K_p

Example First order form of following transfer function

$$\frac{Y(s)}{X(s)} = \frac{2}{s + \frac{1}{3}}$$

Example

First order form of following transfer function

$$\frac{Y(s)}{X(s)} = \frac{2}{s + \frac{1}{3}} = \frac{2}{\frac{3s+1}{3}} = \frac{6}{3s+1}$$

↗ Gain

↓

Time constant
 $\frac{1}{2}$

if $X(s) = \frac{1}{s} \Rightarrow$ unit step function

$$Y(s) = \frac{6}{s(3s+1)}$$

Ultimate value of $Y(t)$ is

$$\lim_{s \rightarrow 0} [Y(s)] = \lim_{s \rightarrow 0} \frac{6}{s(3s+1)} = 6 = K_p \Rightarrow \text{s.s value the system attains after being disturbed}$$

obtained by setting $s=0$ in transfer function

Properties of Transfer Functions

$$\text{Transfer Function} = G(s) = \frac{Y(s)}{X(s)}$$

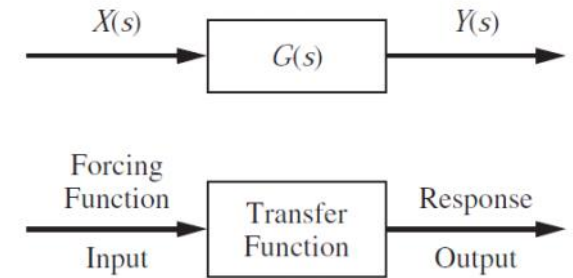
$G(s)$ = *Symbol for transfer function*

$X(s)$ = *Transform of input in deviation form*

(Surrounding temperature of the thermometer)

$Y(s)$ = *Transform of output in deviation form*

(Thermometer reading is the output)



$$Y(s) = G(s)X(s)$$

$$X(s) = a_1X_1(s) + a_2X_2(s)$$

$$Y(s) = G(s)X(s)$$

$$Y(s) = a_1G(s)X_1(s) + a_2G(s)X_2(s)$$

$$Y(s) = a_1Y_1(s) + a_2Y_2(s)$$

Transient Response of forcing functions acting on first-order Transfer function

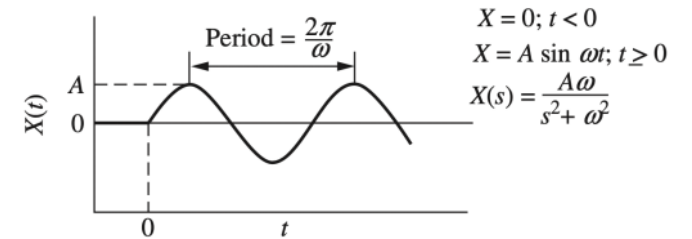
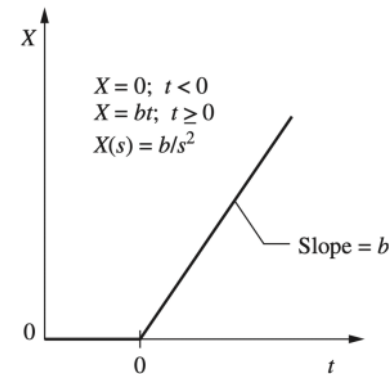
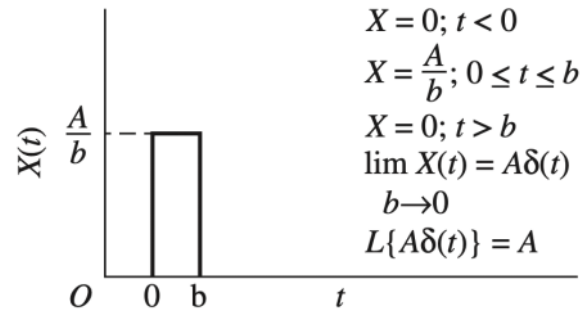
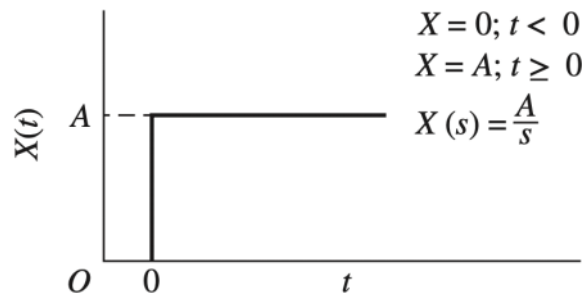
Forcing function types:-

Step

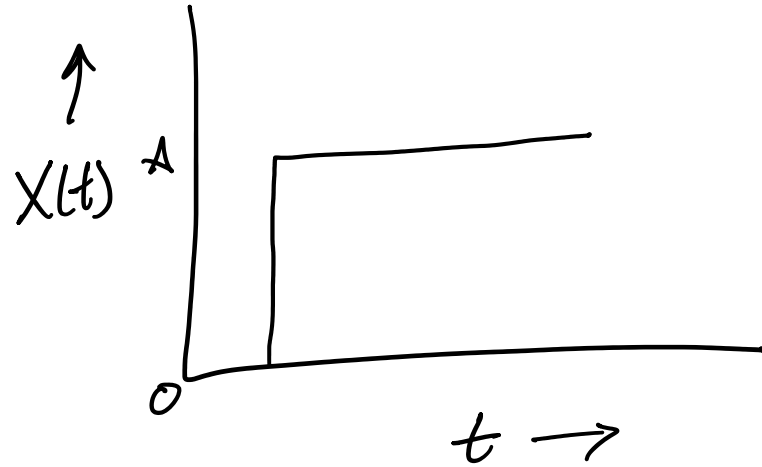
Impulse

Ramp

Sinusoidal



Step function



$$X(t) = 0 \text{ @ } t < 0$$

and

$$X(t) = A \text{ @ } t > 0$$

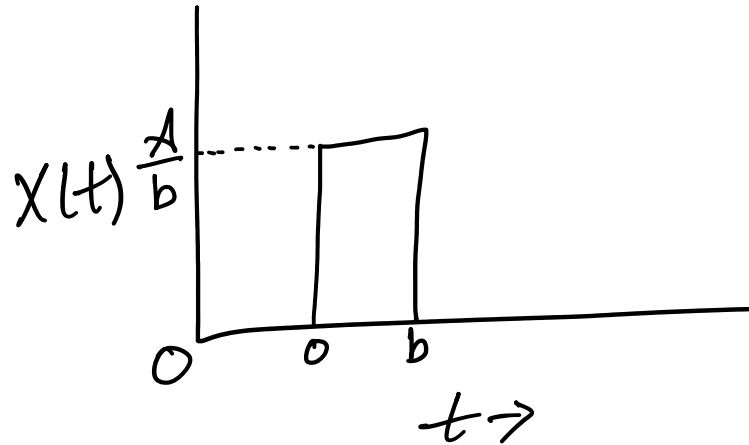
unit step function is $u(t)$
with Laplace transform of $\frac{1}{s}$

so for $X(t) = A u(t)$

$$X(s) = \frac{A}{s}$$

c.g. step change in
flow rate
by
flipping valve open
or shut

Impulse function



For instantaneous function b is close to 0
 $\Rightarrow \lim_{b \rightarrow 0} X(t) = A \delta(t)$
 $\hookrightarrow$ unit impulse

$$\text{and } \mathcal{L} \{A \delta(t)\} = A$$

Same as unit step

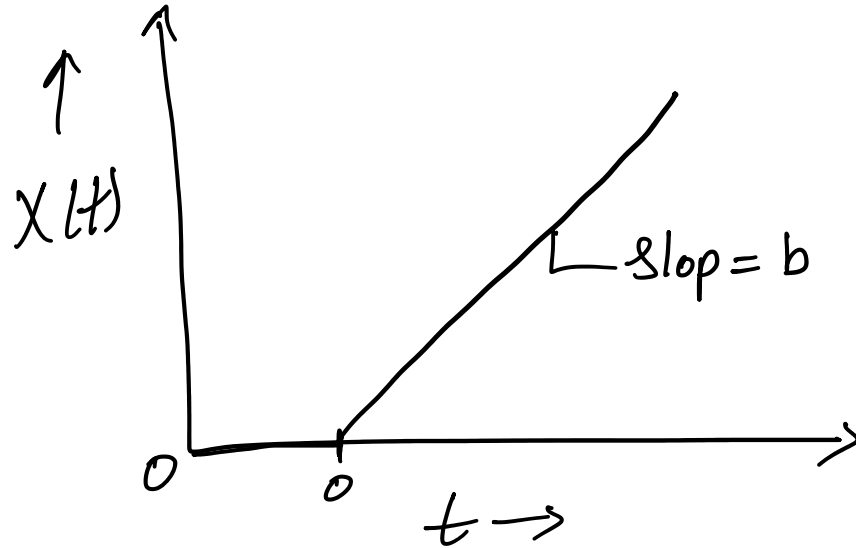
$$X(t) = 0 \quad @ \quad t < 0$$

{ but now @ $t > 0$; $X(t) = \frac{A}{b}$
an additional condition is added
 $t > b$; $X(t) = 0$ }

so between t from 0 to b

$$X = \frac{A}{b} \quad ; \quad 0 \leq t \leq b$$

Ramp Function



e.g of heating or cooling
furnaces or ovens at a fixed
rate

B.C

$$@ t < 0 ; X(t) = 0$$

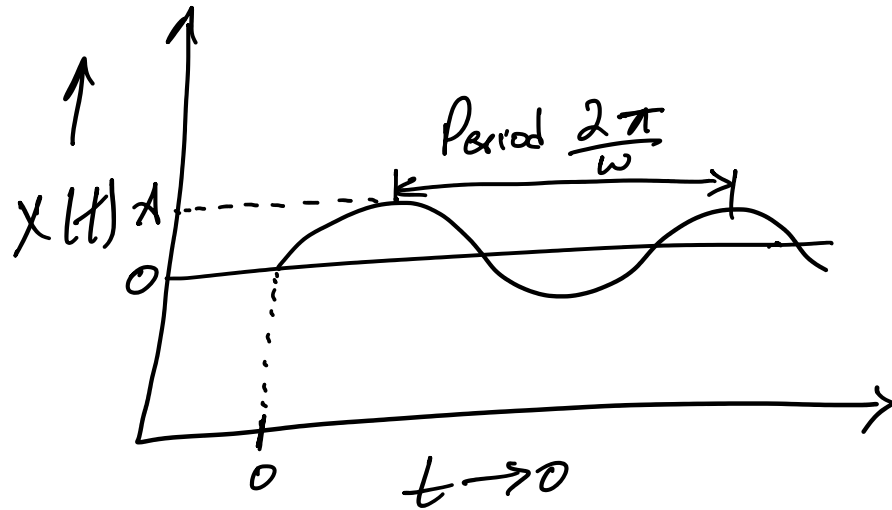
$$@ t \geq 0 ; X(t) = bt$$

and now

$$\mathcal{L}\{X(t)\} = b \mathcal{L}\{t\}$$

$$X(s) = \frac{b}{s^2}$$

Sinusoidal Function



$A = \text{amplitude}$

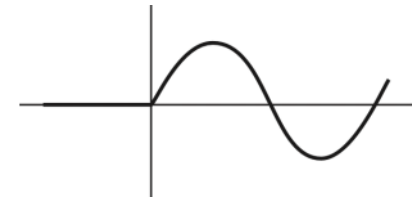
$\omega = \text{frequency} \cdot 2\pi = 2\pi f$

B.C

$$@ t < 0 ; X(t) = 0$$

$$@ t \geq 0 ; X(t) = A \sin(\omega t)$$

$\sin kt u(t)$



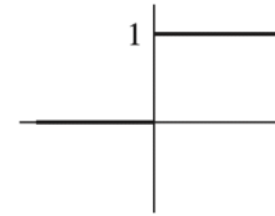
$$\frac{k}{s^2 + k^2}$$

From which we get

$$\mathcal{L}\{X(t)\} = F(s) = \left(\frac{\omega}{s^2 + \omega^2}\right) A$$

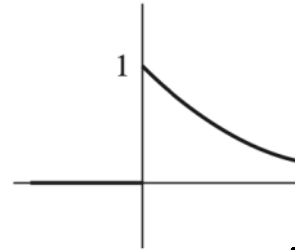
Partial fractions to Fcs $\rightarrow$ time domain

$$X(s) = \frac{2}{s(5s+1)} + \frac{15}{5s+1} \quad u(t)$$



$$\frac{1}{s}$$

$$\begin{aligned} & \frac{2/5}{s(s+1/5)} \quad \frac{15/5}{s+1/5} \\ & \downarrow \quad \downarrow \\ & = \frac{A}{s} + \frac{B}{s+1/5} \quad \frac{3}{s+1/5} \end{aligned} \quad e^{-at}u(t)$$



$$\frac{1}{s+a}$$

$$\frac{2/5}{s+1/5} = A + \frac{Bs}{s+1/5}$$

with $s=0$

$$A = \frac{2/5}{1/5} = 2$$

$$\frac{2/5}{s} = \frac{A(s+1/5)}{s} + B$$

@ $s = -1/5$

$$\frac{2/5}{-1/5} = 0 + B$$

$$B = -2$$

Finally we get

$$= \frac{2}{s} - \frac{2}{s+1/5} + \frac{3}{s+1/5}$$

transforms are

$$= 2 - 2e^{-t/5} + 3e^{-t/5}$$

$$= 2 + e^{-t/5} = X(t)$$

Step Response

Laplace transform for unit step function with magnitude A

$$X(s) = \frac{A}{s}$$

Transfer function of first order systems is

$$\frac{Y(s)}{X(s)} = \frac{1}{\tau s + 1} \Rightarrow \text{Plugging in}$$

$$Y(s) = \frac{A}{s(\tau s + 1)} = \frac{A/\tau}{s(s + 1/\tau)} = \frac{C_1}{s} + \frac{C_2}{s + 1/\tau} = \frac{A}{s} - \frac{A}{s + 1/\tau} = A(1 - e^{-t/\tau})$$

For C_1 $\frac{A/\tau}{s + 1/\tau} = C_1 + \frac{C_2 s}{s + 1/\tau}$

@ $s=0 \Rightarrow C_1 = A$

For C_2 $\frac{A/\tau}{s} = \frac{C_1(s + 1/\tau)}{s} + C_2$

@ $s = -1/\tau$ $C_2 = -A$

For $Y(s) = \frac{A}{s(\tau s + 1)}$

$Y(t) = A(1 - e^{-t/\tau})$

Overall

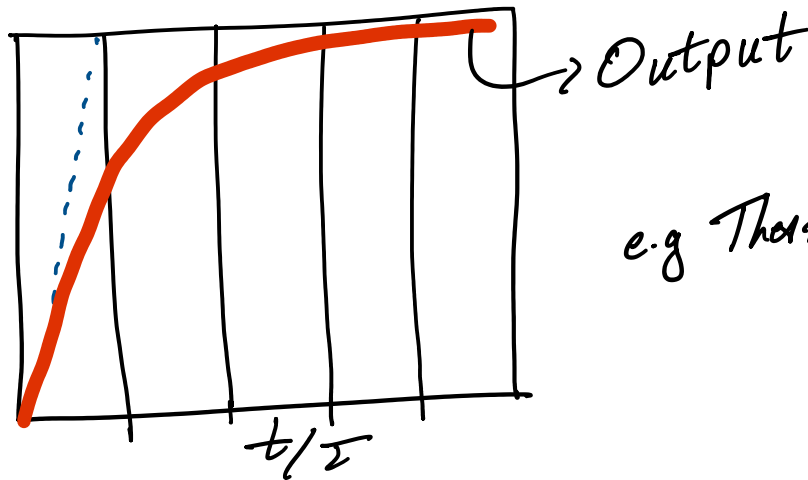
@ $t < 0$; $Y(t) = 0$

@ $t \geq 0$; $Y(t) = A(1 - e^{-t/\tau})$

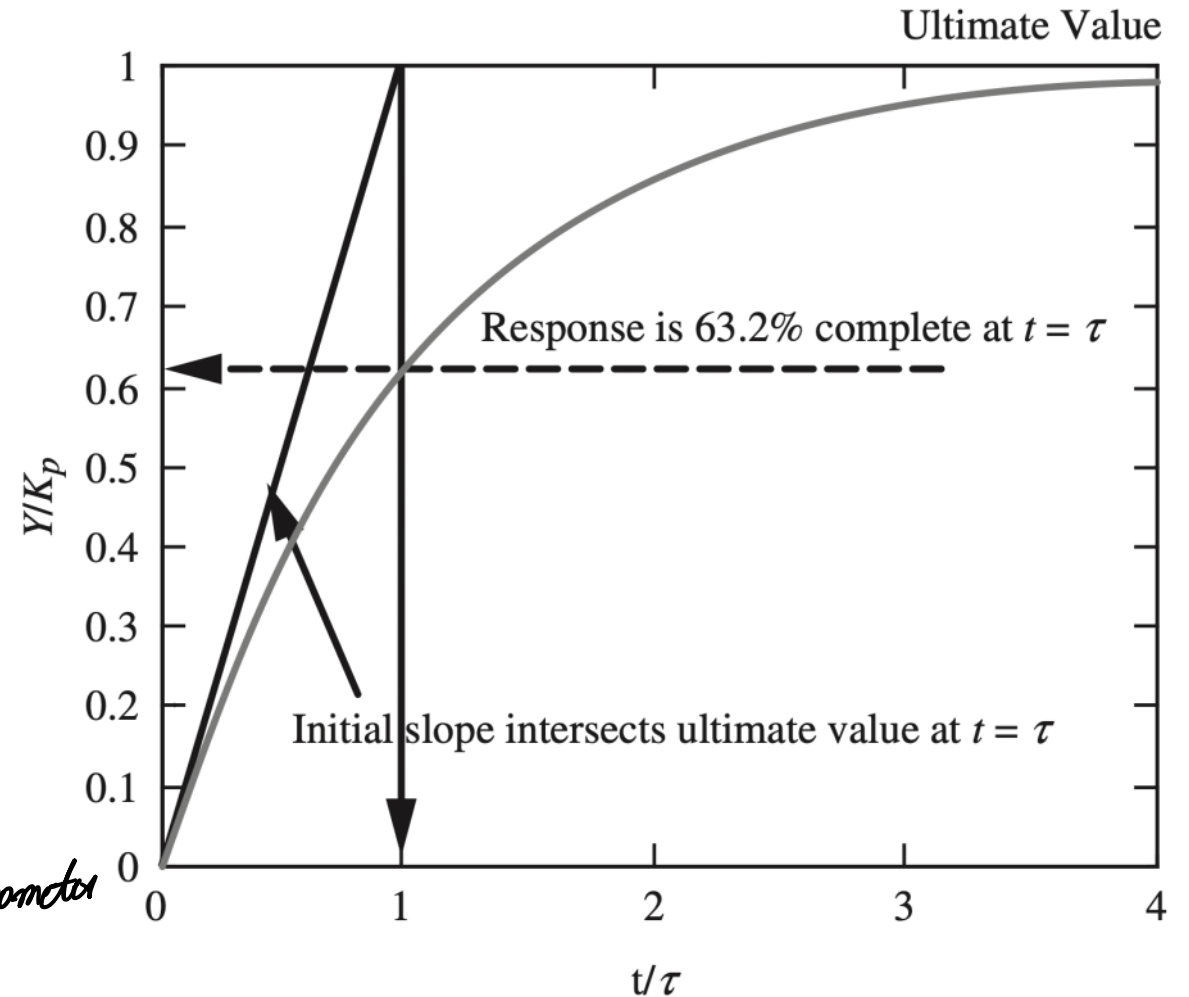
using dimensionless quantities

$\frac{Y(t)}{A}$ vs $\frac{t}{\tau}$

$K_p \Leftarrow \frac{Y(t)}{A}$
Gain



e.g. Thermometer



Example Thermometer with time constant $\tau = 0.1 \text{ min}$

s.s temp = $90^\circ\text{F} = x_s$

@ $t=0$ disturbance temp = 100°F

What is time needed to reach 98°F on thermometer

$\Rightarrow$ The magnitude of disturbance is $A = 100 - 90 = 10^\circ\text{F}$

We use deviation variables so $y(t)$ @ $t=\infty \Rightarrow y(\infty) = 10^\circ\text{F}$
 $= y - y_s$

So for temp of 98°F ; $y(t) = 8^\circ\text{F}$

$$\text{As } y(t) = A(1 - e^{-t/\tau})$$

$$8 = 10(1 - e^{-t/0.1}) \Rightarrow \frac{y}{A} = \frac{8}{10} = 0.8 \Rightarrow \text{for } 80\% \text{ response}$$
$$\frac{t}{\tau} \sim 1.5 \Rightarrow t = 1.5 \times 0.1 = 0.15$$
$$1 - \frac{8}{10} = e^{-t/0.1} ; t = 0.161 \text{ min}$$

Impulse Response

for a unit impulse ; Laplace transform is

$$X(s) = 1$$

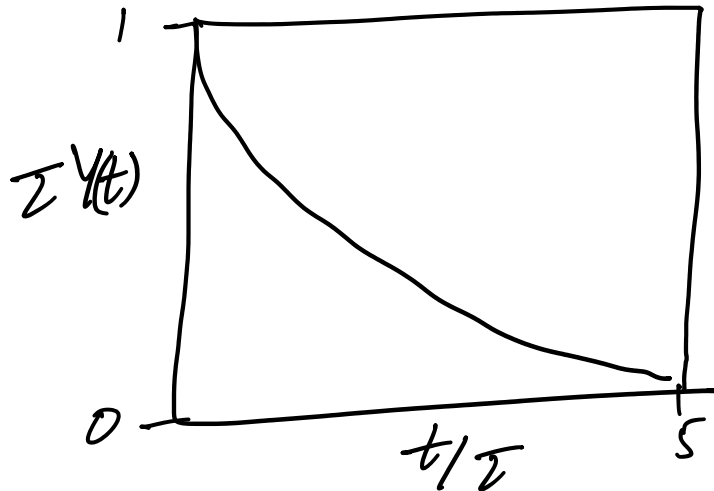
Again adding to transfer function of first order system.

$$\frac{Y(s)}{X(s)} = \frac{K_P}{\tau s + 1}$$

$$\Rightarrow Y(s) = \frac{1}{\tau s + 1} = \frac{1/\tau}{s + 1/\tau}$$

with inverse $y(t) = \frac{1}{\tau} e^{-t/\tau}$

$$\Rightarrow \tau y(t) = e^{-t/\tau}$$



Ramp Response

Ramp input $x(t) = bt \Rightarrow X(s) = b/s^2$

$$Y(s) = \frac{b}{s^2(\tau s + 1)}$$

Rearrange and use partial fractions

$$Y(s) = \frac{b/\tau}{s^2(s + 1/\tau)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s + 1/\tau}$$

$$\begin{aligned} b/\tau &= A s(s + 1/\tau) + B(s + 1/\tau) + C s^2 \\ &= A s^2 + A s/\tau + B s + B/\tau + C s^2 \\ &= (A + C) s^2 + (A/\tau + B) s + B/\tau \end{aligned}$$

For s^2

$$A+C=0 \Rightarrow C=-A$$

For s

$$A/\tau + B = 0 \Rightarrow B = -A/\tau$$

For const.

$$B/\tau = b/\tau$$

$$\boxed{B = b}$$

$\Rightarrow$

$$B = -A/\tau$$

$$b = -A/\tau$$

$$\boxed{A = -b\tau}$$

and $\boxed{C = b\tau}$

$$\frac{b/\tau}{s^2(s+1/\tau)} = \frac{-b\tau}{s} + \frac{b}{s^2} + \frac{b\tau}{s+1/\tau}$$

$$y(t) = b\tau - b\tau(1 - e^{-t/\tau}) = b(t - \tau) + b\tau e^{-t/\tau}$$

Sinusoidal Response

Disturbance of temp $x = x_s + A \sin \omega t \quad t \geq 0$

$$X = x - x_s$$

$$X = A \sin \omega t$$

From transforms

$$X(s) = \frac{A \omega}{s^2 + \omega^2}$$

Plugging into first order T.F

$$Y(s) = \frac{A \omega}{s^2 + \omega^2} \frac{1/2}{s + 1/2}$$

Solve using Partial Fractions

Using transforms

$$\sin kt \, u(t) \rightarrow \frac{k}{s^2 + k^2}$$

$$\cos kt \, u(t) \rightarrow \frac{s}{s^2 + k^2}$$

$$Y(s) = \frac{A\omega}{s^2 + \omega^2} \frac{1/2}{s + 1/2} = \frac{P\omega}{2(s^2 + \omega^2)(s + 1/2)} = \frac{As + B}{s^2 + \omega^2} + \frac{C}{s + 1/2}$$

$$\Rightarrow P\omega = (As + B)2(s + 1/2) + C2(s^2 + \omega^2)$$

$$\frac{P\omega}{2} = As^2 + A\frac{s}{2} + Bs + B\frac{1}{2} + Cs^2 + C\omega^2$$

$$P\frac{\omega}{2} = s^2(A + C) + s(A\frac{1}{2} + B) + (B\frac{1}{2} + C\omega^2)$$

For s^2

$$A + C = 0$$

$$A = -C$$

$$A = -\frac{P\omega/2}{(\frac{1}{2}^2 + \omega^2)}$$

For s

$$A\frac{1}{2} + B = 0$$

$$B = -A\frac{1}{2} \\ = C\frac{1}{2}$$

$$B = \frac{P\omega/2}{(\frac{1}{2}^2 + \omega^2)}$$

Similarly

$$B\frac{1}{2} + C\omega^2 = P\frac{\omega}{2}$$

$$C(\frac{1}{2}^2 + \omega^2) = P\frac{\omega}{2}$$

$$C = \frac{P\omega/2}{(\frac{1}{2}^2 + \omega^2)}$$

$$\frac{Pw}{z(s^2+w^2)(s+\frac{1}{2})} = \left(\frac{-P\frac{w}{2}}{(\frac{1}{2^2}+w^2)} s + \frac{P\frac{w}{2}}{(\frac{1}{2^2}+w^2)} \right) \frac{1}{s^2+w^2} + \frac{P\frac{w}{2}}{(\frac{1}{2^2}+w^2)} \frac{1}{s+\frac{1}{2}}$$

$$= \left(\frac{-Pw\cancel{2}}{1+w^2\cancel{2}^2} s + \frac{Pw}{1+w^2\cancel{2}^2} \right) \frac{1}{s^2+w^2} + \frac{Pw\cancel{2}}{1+w^2\cancel{2}^2} \frac{1}{s+\frac{1}{2}}$$

Inverse Laplace

$$= \frac{-Pw\cancel{2}}{1+w^2\cancel{2}^2} \cos wt + \frac{P}{1+w^2\cancel{2}^2} \sin wt + \frac{Pw\cancel{2}}{1+w^2\cancel{2}^2} e^{-t/2}$$

$$= \frac{Pw\cancel{2}}{w^2\cancel{2}^2+1} e^{-t/2} - \frac{Pw\cancel{2}}{w^2\cancel{2}^2+1} \cos wt + \frac{P}{w^2\cancel{2}^2+1} \sin wt$$

Change Back $P \rightarrow A$

with identity

$$p \cos B + q \sin B = r \sin(B + \theta)$$

$$r = \sqrt{p^2 + q^2} \quad \tan \theta = \frac{p}{q} = -\omega \tau$$

$$r = \sqrt{\left(\frac{-A\omega\tau}{\tau^2\omega^2 + 1}\right)^2 + \left(\frac{A}{\tau^2\omega^2 + 1}\right)^2} = \frac{A}{\sqrt{\tau^2\omega^2 + 1}}$$

$$y(t) = \frac{A\omega\tau}{\tau^2\omega^2 + 1} e^{-\frac{t}{\tau}} + \frac{A}{\sqrt{\tau^2\omega^2 + 1}} \sin(\omega t + \phi)$$

$$\phi = \tan^{-1}(-\omega\tau)$$

$$y(t) \Big|_s = \frac{A}{\sqrt{\tau^2\omega^2 + 1}} \sin(\omega t + \phi)$$

Example $\tau = 0.1 \text{ min}$

Bath temp $= 100^\circ\text{F} = Y_0$

@ $t=0$ variation with amplitude of $2^\circ\text{F} = A$

$$\text{frequency} = \frac{10}{\pi} \text{ cycles/min} = f$$

$$\omega = 2\pi f = 2\pi \frac{10}{\pi} = 20 \text{ rad/min}$$

$$\text{with } Y(t) = \frac{A}{\sqrt{\tau^2 \omega^2 + 1}} = \frac{2}{\sqrt{4 + 1}} = 0.896^\circ\text{F}$$

$$\phi = -\tan^{-1}(-\omega\tau) = -\tan^{-1} 2 = -63.5^\circ = -1.11 \text{ rad}$$

Phase lag $= 63.5^\circ$

$$Y(t) = 0.896 \sin(\omega t + \phi) = 0.896 \sin(20t - 1.11)$$

$$y(t) = 100 + 0.896 \sin(20t - 1.11)$$

A thermometer having a time constant of 0.2 min is placed in a temperature bath, and after the thermometer comes to equilibrium with the bath, the temperature of the bath is increased linearly with time at a rate of $1^\circ/\text{min}$. Find the difference between the indicated temperature and the bath temperature.

$$\tau = 0.2 \text{ min}$$

$$\text{Ramp} = 1^\circ/\text{min} = b$$

Diff in bath temp x and indicated temp y @ 0.1 min after change in temp begins

For Ramp Input

$$\begin{aligned} Y(t) &= bt - b\tau(1 - e^{-t/\tau}) \\ &= b(t - \tau) + b\tau e^{-t/\tau} \end{aligned}$$

$$@ t = 0.1 \text{ min}$$

$$Y(t) = 1(0.1 - 0.2) + 1(0.2)e^{-\frac{0.1}{0.2}} = -0.1 + 0.121 = 0.021^\circ\text{F}$$

$$@ 0.1 \text{ min}$$

$$X(t) = 0.1$$

$$\begin{aligned} X(t) - Y(t) \\ = 0.079 \end{aligned}$$

@ $t = 1.0 \text{ min}$

$$Y(t) = 1(1 - 0.2) + 1(0.2)e^{-\frac{1}{0.2}} = 0.8 + 1.34 \times 10^{-3} = 0.801^\circ\text{F}$$

$$@ 1 \text{ min } X(t) = 1^\circ\text{F}$$

$$\begin{aligned} X(t) - Y(t) &= 1 - 0.801 \\ &= 0.199 \end{aligned}$$

Homework 3

Name: _____

Construct a block diagram showing the response of a first order system to a step function with gain and time constant as unity.

- Test the response for higher and lower values of time constant. Explain what changes occur.

Note: Please use your Registration number as the gain of the system. Prepare the model in Simulink and present your results concisely in a printed format. Do not submit the Simulink file.

Name: _____

A thermometer having first-order dynamics with a time constant of 1 min is placed in a temperature bath at 100°F . After the thermometer reaches steady state, it is suddenly placed in a bath at 110°F at $t = 0$ and left there for 1 min, after which it is immediately returned to the bath at 100°F .

- (a) Draw a sketch showing the variation of the thermometer reading with time.
- (b) Calculate the thermometer reading at $t = 0.5$ min and at $t = 2.0$ min.

Tips:

For pasting pictures, try taking screenshots using "Windows+Shift+S" and pasting directly on here
For adding word or pdf documents, add them here and right click to select "Insert as Printout"